

A Technical Introduction to Medical image computing

Section 1

Shape analysis Shape analysis is the field of medical image computing that studies geometrical properties of structures obtained from different imaging modalities. Shape analysis recently become of increasing interest to the medical community due to its potential to precisely locate morphological changes between different populations of structures, i.e. healthy vs pathological, female vs male, young vs elderly. Shape analysis includes two main steps: shape correspondence and statistical analysis.

Section 2

Image registration is a process that searches for the correct alignment of images. In the simplest case, two images are aligned. Typically, one image is treated as the target image and the other is treated as a source image; the source image is transformed to match the target image. The optimization procedure updates the transformation of the source image based on a similarity value that evaluates the current quality of the alignment. This iterative procedure is repeated until a (local) optimum is found. An example is the registration of CT and PET images to combine structural and metabolic information (see figure). Image registration is used in a variety of medical applications:

Section 3

Shape correspondence is the methodology that computes correspondent locations between geometric shapes represented by triangle meshes, contours, point sets or volumetric images. Obviously definition of correspondence will influence directly the analysis. Among the different options for correspondence frameworks are: anatomical correspondence, manual landmarks, functional correspondence (i.e. in brain morphometry locus responsible for same neuronal functionality), geometry correspondence, (for image volumes) intensity similarity, etc. Some approaches, e.g. spectral shape analysis, do not require correspondence but compare shape descriptors directly. Statistical analysis will provide measurements of structural change at correspondent locations.

Section 4

Data forms Medical image computing typically operates on uniformly sampled data with regular x-y-z spatial spacing (images in 2D and volumes in 3D, generically referred to as images). At each sample point, data is commonly represented in integral form such as signed and unsigned short (16-bit), although forms from unsigned char (8-bit) to 32-bit float are not uncommon. The particular meaning of the data at the sample point depends on modality: for

example a CT acquisition collects radiodensity values, while an MRI acquisition may collect T1 or T2-weighted images. Longitudinal, time-varying acquisitions may or may not acquire images with regular time steps. Fan-like images due to modalities such as curved-array ultrasound are also common and require different representational and algorithmic techniques to process. Other data forms include sheared images due to gantry tilt during acquisition; and unstructured meshes, such as hexahedral and tetrahedral forms, which are used in advanced biomechanical analysis (e.g., tissue deformation, vascular transport, bone implants).

Section 5

Studying temporal changes. Longitudinal studies acquire images over several months or years to study long-term processes, such as disease progression. Time series correspond to images acquired within the same session (seconds or minutes). They can be used to study cognitive processes, heart deformations and respiration. Combining complementary information from different imaging modalities. An example is the fusion of anatomical and functional information. Since the size and shape of structures vary across modalities, it is more challenging to evaluate the alignment quality. This has led to the use of similarity measures such as mutual information. **Characterizing a population of subjects.** In contrast to intra-subject registration, a one-to-one mapping may not exist between subjects, depending on the structural variability of the organ of interest. Inter-subject registration is required for atlas construction in computational anatomy. Here, the objective is to statistically model the anatomy of organs across subjects. **Computer-assisted surgery.** In computer-assisted surgery pre-operative images such as CT or MRI are registered to intra-operative images or tracking systems to facilitate image guidance or navigation. There are several important considerations when performing image registration:

Section 6

Small sample size (curse of dimensionality): a large medical imaging dataset contains hundreds to thousands of images, whereas the number of voxels in a typical volumetric image can easily go beyond millions. A remedy to this problem is to reduce the number of features in an informative sense (see dimensionality reduction). Several unsupervised and semi-/supervised, approaches have been proposed to address this issue. **Interpretability:** A good generalization accuracy is not always the primary objective, as clinicians would like to understand which parts of anatomy are affected by the disease. Therefore, interpretability of the results is very important; methods that ignore the image structure are not favored. Alternative methods based on feature selection have been proposed,.

Section 7

Massive univariate approaches that probe individual voxels in the imaging data for a relationship to the experiment condition. The prime approach is the general linear model. Multivariate- and classifier based approaches, often referred to as multi voxel pattern analysis or multi-variate pattern analysis probe the data for global and potentially distributed responses to an experimental condition. Early approaches used support vector machines to study responses to visual stimuli. Recently, alternative pattern recognition algorithms have been explored, such as random forest based gini contrast or sparse regression and dictionary learning. Functional connectivity analysis studies the intrinsic network structure of the brain, including the interactions between regions. The majority of such studies focus on resting state data to parcellate the brain or to find correlates to behavioral measures. Task specific data can be used to study causal relationships among brain regions (e.g., dynamic causal mapping). When working with large cohorts of subjects, the normalization (registration) of individual subjects into a common reference frame is crucial. A body of work and tools exist to perform normalization based on anatomy (FSL, FreeSurfer, SPM). Alignment taking spatial variability across subjects into account is a more recent line of work. Examples are the alignment of the cortex based on fMRI signal correlation, the alignment based on the global functional connectivity structure both in task-, or resting state data, and the alignment based on stimulus specific activation profiles of individual voxels.

Section 8

Group analysis In the group analysis, the objective is to detect and quantize abnormalities induced by a disease by comparing the images of two or more cohorts. Usually one of these cohorts consist of normal (control) subjects, and the other one consists of abnormal patients. Variation caused by the disease can manifest itself as abnormal deformation of anatomy (see voxel-based morphometry). For example, shrinkage of sub-cortical tissues such as the hippocampus in brain may be linked to Alzheimer's disease. Additionally, changes in biochemical (functional) activity can be observed using imaging modalities such as positron emission tomography. The comparison between groups is usually conducted on the voxel level. Hence, the most popular pre-processing pipeline, particularly in neuroimaging, transforms all of the images in a dataset to a common coordinate frame via medical image registration in order to maintain correspondence between voxels. Given this voxel-wise correspondence, the most common frequentist method is to extract a statistic for each voxel (for example, the mean voxel intensity for each group) and perform statistical hypothesis testing to evaluate whether a null hypothesis is or is not supported. The null hypothesis typically assumes that the two cohorts are drawn from the same distribution, and hence, should have the same statistical properties (for example, the mean values of two

groups are equal for a particular voxel). Since medical images contain large numbers of voxels, the issue of multiple comparison needs to be addressed,. There are also Bayesian approaches to tackle group analysis problem.